

Enterprise Digital Infrastructure

Definitions & Basic Concepts

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Enterprise digital infrastructure is the complex digital ecosystem that is the core of the digital economy and of the supply chain of services and products

This ecosystem includes

- **technological** components:

hardware components: servers, storage devices, backup devices, tape libraries, GPUs, FPGAs, sensors, actuators, ...

software components: operating systems, DBMS, web applications, HPC applications, gaming applications, ...

communication/networking components: cables, antennas, satellites, routers, access points, base stations, ...

- **non-technological** components:

power distribution systems, UPS, cooling systems, HVAC, smoke detectors, fire extinguishers, anti-intrusion systems, CCTVs, ...

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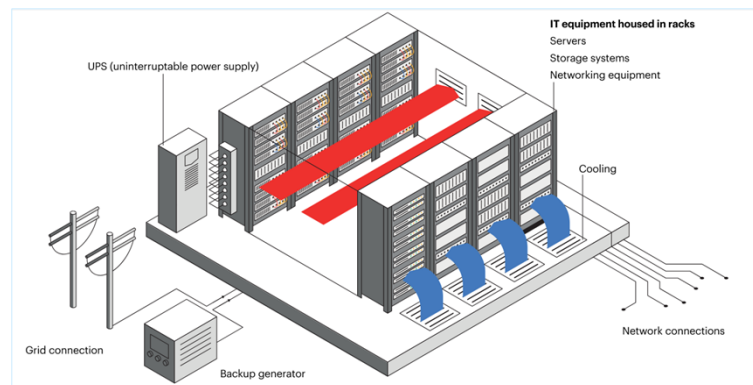
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Data centers

Data centers are the “*smart buildings*” housing this infrastructure

Backbone of the inter-connected world that facilitates the seamless flow of data



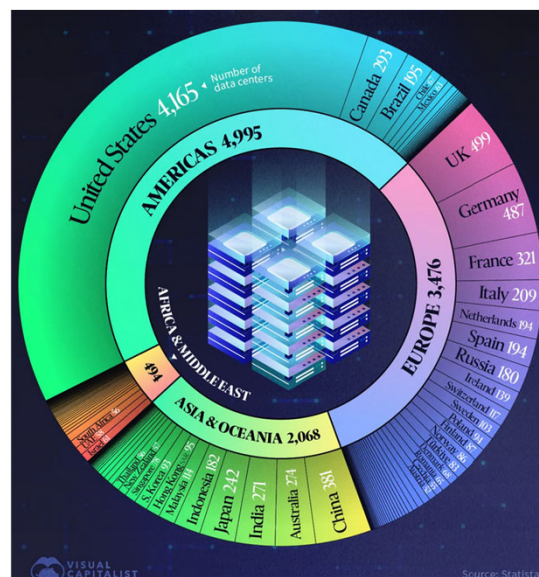
Source: International Energy Agency

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Distribution of data centers [Nov 2025]



Source: Statista

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Data centers (cont'd)

Data centers are owned and operated by companies/organizations for

- Running enterprise applications
- Deploying services
- Renting computing/storage resources
- Providing co-location services

Data centers are classified in *tiers* according to their reliability [*standard by Uptime Institute*]

Public or private data centers

Services and computing/storage resources are accessible over the Internet

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Data centers & electricity

Data centers are energy-hungry

Electricity consumption is growing rapidly due to energy-intensive accelerated computing mainly used for AI purposes



12% growth per year over the last 5 years

Projections indicate that it will more than double towards 945 TWh by 2030

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Data center sustainability

United Nations defined sustainability as *"meeting the needs of the present without compromising the ability of future generations to meet their own need"*

To be sustainable, data centers should be environment-friendly in terms of carbon footprint, water usage and waste production

Energy efficiency involves advanced cooling systems, efficient server architectures and intelligent power distribution systems

Power Usage Effectiveness (PUE) measures energy efficiency as ratio between power entering data center and power used to run digital infrastructure

The smaller PUE better

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Recap

Enterprise digital infrastructure

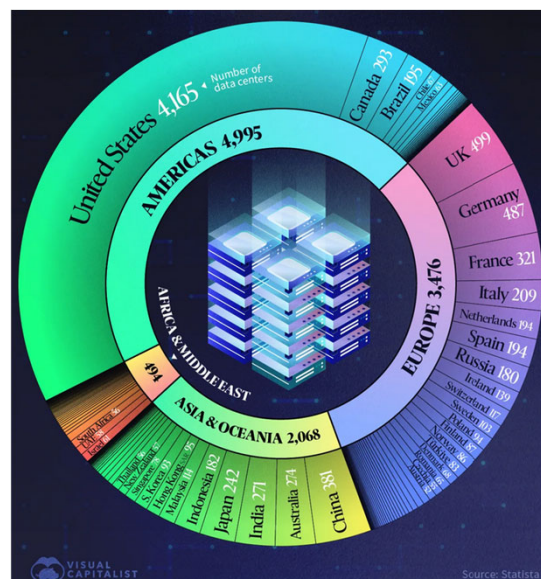
Data centers & sustainability

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Distribution of data centers [Nov 2025]

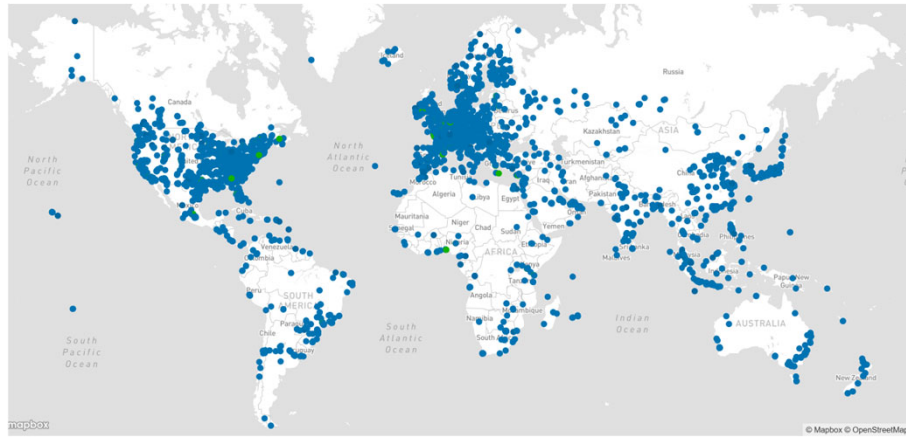


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Data center map



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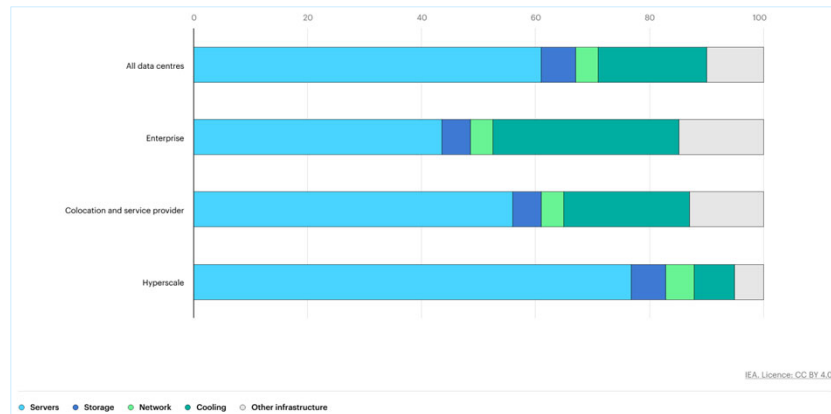
The smaller PUE better

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Share of electricity by data center type



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Data centers & renewal energy sources

Solar and wind power offer significant environmental advantages at low cost with respect to fossil fuels

Renewal energy sources lead to carbon neutral data centers

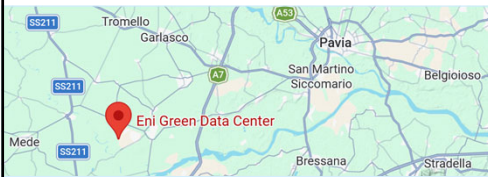


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ENI Green Data Center



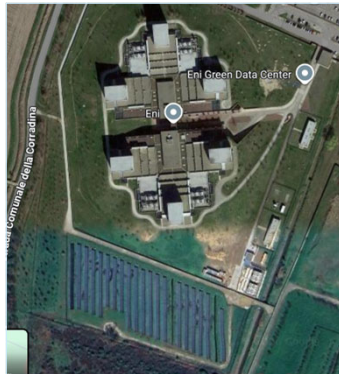
ENI data center hosts the HPC6 super-computer, consisting of about 3,500 nodes with 1 CPU and 4 GPUs each

606 Pflops peak computational power

Maximum power consumption ~10MW with a solar plant providing ~1MW

Natural “free-cooling” system

Liquid cooling system able to dissipate 96% of the heat generated by machines



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AI-based services impacts

ChatGPT receives about 2.5 billions prompts per day

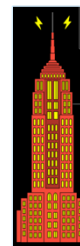
Each prompt uses an estimated 0.34 watt-hours of electricity (ten times higher than a Google query) with a total of 680,000 kilowatt-hours daily

ChatGPT Prompts Consumes Enough Electricity Daily to Power the Empire State Building for a Year and a Half

Each month ChatGPT produces more than 260,000 kilograms of CO₂

ChatGPT produces the same amount of CO₂ emissions as 260 flights from New York City to London each month, study finds

By Tech Brew and Tricia Crimmins
January 21, 2025, 9:04 AM ET



The Guardian

AI boom has caused same CO₂ emissions in 2025 as New York City, report claims

Robert Booth UK
technology editor
Thu 18 Dec 2025 12:15 CET

Study author says tech companies are reaping benefits of artificial intelligence age but society is left to pay cost

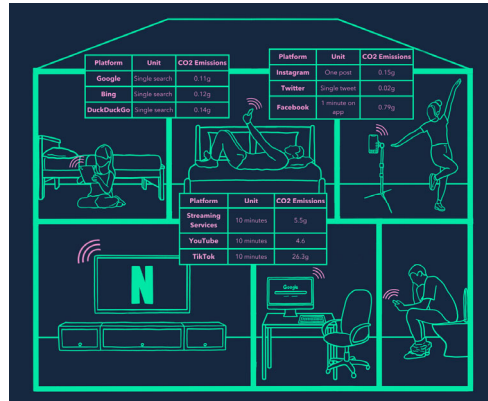
‘Just an unbelievable amount of pollution’: how big a threat is AI to the climate?

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Emissions from digital technologies



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Choice of the infrastructure

The choice of the “right” technological components is very challenging:

- ingredients vs recipe
- intuition

The choice depends on:

- ✓ type of service/application
- ✓ workload/traffic
- ✓ quality/performance requirements
- ✓ cost

Two contradictory requirements:

- quality → efficiency → **over provisioning**
- cost → competitiveness → **under provisioning**

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Performance challenges

Services are *on-demand* (utility computing)

Service *popularity* varies with time due to their social nature and to external events

- Network effect
- Flash crowd
- Slashdot effect

- Limited-edition product or sneaker drops
- Large flash sales (e.g. limited-time offers)
- Ticket onsales for a popular artist or festival
- High-profile public sector registrations

Location-based/context-aware services use location data to customize the content

Services deployed on *geographically distributed* (virtualized) infrastructure

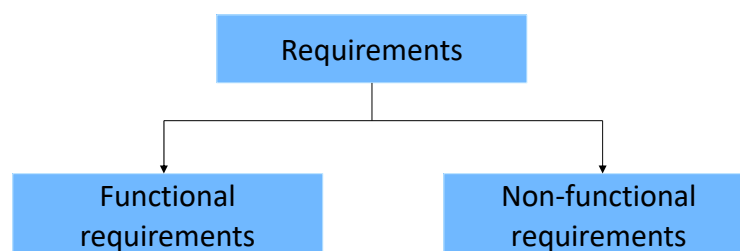
Composition of multiple services dynamically selected and of heterogeneous data flows under the control of different providers

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System/service requirements



Functional requirements define *what* a system/service should do, i.e., specific features and operations it must perform to meet business and user needs

Non-functional requirements define *how* a system/service should operate by focusing on user experience rather than on specific features

No direct relationship: a system/service can still work if non-functional requirements are not met, although it might not satisfy user expectation or business needs

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Quality of Service (QoS)

Non functional requirements are very important for an effective digital transformation

Quality of Service (QoS) refers to *non-functional requirements* related to system/service behavior and user experience 😊 😡

- Dependability
 - Reliability (uptime and downtime)
 - Trustworthiness (privacy and security)
- Performance
- Usability

Resilience refers to the ability to maintain essential operational capabilities in the face of adversity (e.g., flash crowd, security attack, power outage) and rapidly recover to an effective operational posture [Source: NIST]

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Properties of QoS attributes

Very *subjective* attributes: perception of the users depends on many factors, e.g., their type/skill/age, service type, level of interactivity, timing requirements

Concept of *patience*: ability to accept delays without getting upset



How long users can wait? Some numbers: 0.1 sec “instantaneous” response, 1 sec still ok, 10 sec so-so, >10 sec too much

Mobile users and generation Z are not very patient



Patience loss leads to **frustration**

Lack/loss of patience significantly affects online businesses, e.g., loss of customers, drop in sales/revenues, reputation damages

Waiting times should be taken into account on the design phase

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Usability

Broad quality attribute including many aspects of a service

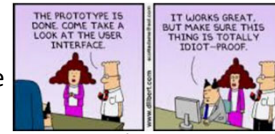
It measures the experience of the users in interacting with systems/services
(*user satisfaction*) 👍 👎

It is defined as the ability to **learn** and **use** a given system/service and **memorize** it

Very subjective

It is expressed in terms of

- *efficiency*: ability to complete a task easily, quickly and without frustration
- *effectiveness*: ability to successfully complete a task (and obtain the desired results)



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Usability (cont'd)

The usability depends on

- the design of the user interface (UI)
- the technological infrastructure and its performance

Some services (e.g., video/audio services, VoIP) might become unusable because of poor performance of the technological infrastructure on the provider side (and also on the user side)

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Quality of Service (QoS)

Quality of Service (QoS) attributes

- Dependability
 - Reliability (uptime and downtime)
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- Performance
- Usability

Resilience is the *persistence* of dependability when facing *changes*

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Reliability

Reliability is the ability of a given (hardware or software) component to perform/work as planned (i.e., continuity of correct behavior)

Reliability includes the concept of *availability*

Availability is the quality of a system/service of being reachable and working properly whenever requested (i.e., readiness)

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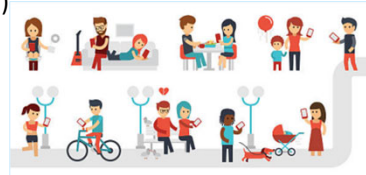
Availability

Availability defined as the fraction of time during which a system/service is reachable and works properly

$$\text{availability} = \text{uptime} / (\text{uptime} + \text{downtime})$$

The ideal case 24/7 (=100%) availability is desirable but unrealistic

Downtimes noticed immediately (we are addicted)



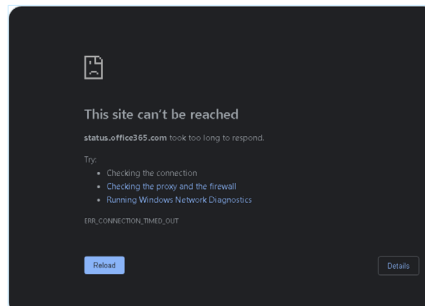
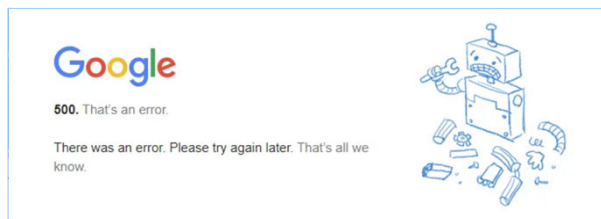
The tolerance of the users to unavailability is very *subjective*

Unavailability may cause many inconveniencies, inefficiencies, damages to reputation, loss of trust, financial losses, ...

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